Sujoy Moulick

Full-Stack Developer | Graphic Designer- ACM Student Chapter , UEMJ
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Location : Jaipur, Rajasthan

Professional Summary

Full-Stack Developer & SaaS Architect with extensive experience in building interactive gaming platforms and client-side web applications. Proficient in frontend architecture, real-time database integration with Supabase, and production-level authentication systems. Skilled in 3D/2D game logic, dynamic DOM manipulation, and full-stack deployment using modern JavaScript frameworks and cloud infrastructure.

Education

Bachelor of Technology in AI & Machine Learning | Aug 2024 – June 2028 University of Engineering & Management (UEM) | Jaipur, Rajasthan Pursuing – 2nd Year

Technical Skills

- Artificial Intelligence & Machine Learning: Skilled in implementing supervised and unsupervised learning algorithms, including Clustering, Regression, and Neural Networks. Proficient in data preprocessing and model evaluation using Python libraries.
- Web Architecture & SaaS Development: Experienced in building full-stack platforms (VectorX) with integrated 3D/2D game logic. Proficient in managing production-level environments, including real-time synchronization and scalable service structures.
- Frontend Engineering & UI Design: Expert in React, Tailwind CSS, and JavaScript for creating dynamic, responsive user interfaces. Capable of building custom client-side tools like TEXTORA with integrated AI-driven formatting.
- Database & Cloud Management: Proficient in Supabase (PostgreSQL), MongoDB, and MySQL for relational and non-relational data handling. Experienced in configuring Google OAuth and managing production-level user access.
- Programming Languages: Well-versed in C, C++, Java, and Python for algorithmic problem solving and software development. Experienced in using MATLAB for technical computing and data analysis.
- Software & Tool Proficiency: Expert in GitHub (Version Control, PR Management, Branching), Adobe Photoshop for graphic assets, and DNS management via GoDaddy for production deployment.
- API & Real-time Integration: Capable of integrating RESTful APIs for live data fetching (Weather/Meghdoot) and implementing secure backend logic using Node.js and Express.
- Development Standards & Optimization: Well-versed in ES6+ standards, clean code principles, and cross-browser optimization. Experienced in performing DOM updates and component-level performance checks.
- Project Documentation: Prepare technical READMEs, structural portfolio galleries, and calculation-based logic sheets to ensure project transparency and code maintainability.
- App Development: Experienced in building cross-platform mobile applications using Flutter, focusing on performance and material design principles.

Project Details

- VectorX Gaming SaaS Simulator | Full-Stack Platform
  - Architecture & Logic: Engineered a scalable gaming ecosystem featuring a library of 20+ retro-themed 3D and 2D games using modern JavaScript engines.
  - Database Integration: Implemented Supabase (PostgreSQL) for real-time player data synchronization, state management, and high-concurrency leaderboard handling.
  - Authentication: Configured production-level Google OAuth and managed secure user access protocols for global deployment.
  - Physics Engine: Developed custom "Antigravity" mechanics and collision detection logic to enhance gameplay interactivity.**Deployed URL**: https://vector-x-eight.vercel.app/

- TEXTORA | Client-Side Web Application
  - Functional Design: Built a lightweight, browser-based document processor for real-time formatting and high-fidelity PDF export.
  - AI Integration: Leveraged AI-driven tools to automate professional text styling and layout optimization.
  - UI/UX Development: Designed a modern, transparent user interface utilizing advanced CSS glassmorphism and responsive design principles.
- MEGHDOOT | Real-Time Weather Utility
  - API Implementation: Architected a weather tracking system using RESTful APIs to fetch global meteorological data with a 7-day forecast capability.
  - Frontend Stack: Developed the interface using Tailwind CSS and vanilla JavaScript, focusing on minimal load times and mobile-first responsiveness.
  - Data Validation: Implemented error-handling logic for location-based search queries and unit conversion (Metric/Imperial).
- ADHYAYAN | Gamified Learning Platform
  - Conceptual Framework: Designed a frontend-heavy learning portal focused on "Learn, Compete & Grow" mechanics to increase student engagement.
  - Interaction Design: Developed interactive modules and achievement-tracking UI components to visualize user progress dynamically. **Deployed URL**: https://viteadhyayan.vercel.app/
- Technical Portfolio & C-Program Repo | Performance Showcase
  - Portfolio Management: Deployed a centralized digital portfolio on GitHub Pages, managing custom DNS records via GoDaddy to ensure zero-downtime availability.
  - Algorithmic Repository: Curated an extensive library of C-based solutions focusing on Data Structures (DSA), recursion, and memory optimization.
  - DevOps Workflow: Utilized advanced GitHub actions, branch management, and pull request reviews to maintain repository code quality.
- FINDO | Personal Finance & Expense Tracker
  - Data Management & Persistence: Developed a secure client-side storage architecture using Local Storage and IndexedDB to ensure offline-first data persistence without external server dependency.
  - Dynamic Financial Logic: Engineered automated calculation engines to track daily expenditures, categorize income streams, and visualize budget health through real-time balance updates.
  - Interactive Data Visualization: Integrated Chart.js to generate dynamic spending reports, providing users with high-fidelity visual analysis of category-wise expense distribution.
  - UI/UX Optimization: Implemented a modular interface with custom form validation, error handling, and responsive layouts to ensure a seamless experience across desktop and mobile platforms.
  - Security & Privacy: Designed a local-only data protocol, ensuring 100% user privacy by preventing sensitive financial data from being transmitted to external servers. **Deployed URL**: https://findo-ten.vercel.app/

TECHNICAL PORTFOLIO | Digital Infrastructure & Deployment

- CI/CD Pipeline: Developed a streamlined deployment workflow utilizing Vercel, which allows for automated production builds and instant global edge distribution with each update to the GitHub repository.
- DNS & Domain Management: Set up custom domain routing through GoDaddy; effectively managed DNS zone files to resolve "Parked" status and create a secure, production-ready SSL environment.
- Frontend Architecture: Crafted a modular, high-performance interface using HTML5, Tailwind CSS, and JavaScript, featuring a dynamic DOM-based navigation system and optimized asset loading for faster Largest Contentful Paint (LCP).
- Version Control: Established a thorough Git workflow, overseeing branch protection, pull requests, and repository synchronization to guarantee zero-downtime during production updates.
- Visual Engineering: Designed and incorporated high-fidelity UI elements, including custom-coded interactive icons, glassmorphic modals, and data-driven visualizations for the "Skill Profile."

Deployed URL: https://portfoli0-amber.vercel.app/

Work Experience (Independent Projects)

Full-Stack Web Developer Self-Employed / Freelance | July 2024 – Present

- Developed and deployed scalable SaaS platforms and client-side utilities using Vercel and GitHub.
- Architected real-time database schemas and authentication flows using Supabase and Google OAuth.
- Engineered custom physics engines and game logic for 20+ retro-themed 3D/2D interactive modules.
- Implemented local-first data persistence architectures using IndexedDB and Local Storage for privacy.
- Managed DNS records and SSL certification via GoDaddy to establish production-ready web environments.
- Optimized web performance metrics (LCP/FID) by implementing modular code and asset lazy-loading.
- Facilitated version control through advanced Git workflows, including branch protection and PR management.

KEY ACHIEVEMENTS

- Successfully deployed a production-grade Gaming SaaS (VectorX) with integrated cloud authentication.
- Developed a high-performance client-side PDF export tool (TEXTORA) featuring AI-driven formatting.
- Orchestrated a CI/CD pipeline via Vercel for automated builds and global edge distribution.
- Created a comprehensive technical portfolio with a 99.9% uptime record and custom domain routing.
- Optimized financial tracking logic in FINDO to ensure sub-second local data processing.